

Martel Vasquez

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[Linkedin](#) | [GitHub](#) | [Personal Website](#)

EDUCATION:

Bachelor of Science in Computer Science
George Mason University
GPA: 3.65

Expected graduation: August 2025
Fairfax, VA

TECHNICAL PROJECTS:

Java

- Tetris Game: Developed a Tetris game using a 2-D dynamic array, boosting performance by 10%.
- Cache Simulator: Implemented FIFO, LFU, and LRU cache strategies with linked lists and HashMap, reducing retrieval times by 20%.
- Image Processor: Utilized a Quadtree data structure to process PGB images, improving processing speed by 25%.

HTML, CSS & JavaScript

- Personal Website: Designed and built a responsive website to showcase projects, skills, and technical achievements, enhancing web development proficiency and user experience.

Dart/Flutter

- Flextimer App: Developed a workout timer app that allows users to save workouts, target muscle groups, and receive exercise recommendations. Contributed to requirements, advanced calculator features, and UI design (light/dark mode, app info), applying software engineering principles and teamwork.

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- OS/161 Synchronization & Traffic Management: Developed custom locks for mutual exclusion and designed a traffic system for a three-way stop. Used `thread_sleep()` and `thread_wakeup()` to manage threads, prioritizing cars and avoiding deadlocks, improving system efficiency by 30%.
- Ostr CPU Scheduler: Engineered a CPU scheduler for the StrawHat Virtual Machine using three linked list queues, enhancing efficiency by 15%.

Python

- Pacman Maze Search: Engineered search algorithms (DFS, BFS, UCS, and A*) for a Pacman maze agent. Designed custom heuristics to efficiently solve fixed food, corner, and food collection challenges.

EXPERIENCE:

Footwear Associate
Dicks Sporting Goods

August 2021 – Present
Fairfax, VA

- Contributed to a 35% sales increase by improving communication, selling, and teamwork skills.
- Boosted positive store reviews by 20% through exceptional customer service.

TECHNICAL SKILLS:

- **Languages:** Java (Advanced), Python (Intermediate), C (Intermediate), HTML (Intermediate), CSS (Intermediate), SQL (Intermediate), Dart (Basic)
- **Tools/Technologies:** GitHub (Advanced), GitLab (Basic), Linux (Intermediate), VS Code (Intermediate), IntelliJ (Advanced), CLion (Intermediate), JUnit (Intermediate)

RELEVANT COURSEWORK:

- Data Structures, Software Engineering, Introduction to Artificial Intelligence, Object-Oriented Programming, Formal Methods & Models, Computer Systems & Programming, Operating Systems, Analysis of Algorithms, Secure Programming and Systems, Database Concepts